33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10004

F13-SR-004
415V RECYCLE CRUSHING SWITCHROOM

F13-TR-014
TRANSFORMER ROOM

F13-TR-014
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F13-SW-014
RECYCLE CRUSHING 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000 /5A
5VA
5P20
4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
ACB
AFD
50G
4000A
50
51
24VDC CONTROL POWER SUPPLY
NOTE 7
4000 /5A
50VA
C11.0
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 4000A 65kA for 1sec

1F-05 20A 70A EOL
1R-01 1000A 100/1A
2F-01 1000A 100/1A
1F-09 800A 1000A 100/1A
5F-01 250A 100/1A
5F-05 160A 100/1A
INC
3R-17 400A 700A
FEEDER TO F13-SW-114
2000-F13-SLD-EL-10411
2000-F13-SCM-EL-10442
3R-08 250A AHF
5F-13 100A 90A EOL
4R-13 630A 100/1A
5R-13 630A 100/1A
E-SHUT

CONTINUED ON SHT 2
2000-F13-SLD-EL-10402

M 3~ 5.5kW
F13-BLC-401
SAG MILL
BALL CHARGER
2000-F13-SCM-EL-10408

RTD M 3~ 375kW
F13-CC-401
RECYCLE CRUSHER C
2000-F13-SCM-EL-10409

RTD M 3~ 375kW
F13-CC-402
RECYCLE CRUSHER D
2000-F13-SCM-EL-10410

RTD M 3~ 355kW
F13-CV-403
SAG MILL
FEED CONVEYOR
2000-F13-SCM-EL-10435

t° M 3~ 75kW
F13-CV-405
RECYCLE CONVEYOR 2
2000-F13-SCM-EL-10412

t° M 3~ 45kW
F13-CV-406
RECYCLE CRUSHER
DISCHARGE CONVEYOR
2000-F13-SCM-EL-10413

F13-SW-114
RECYCLE CRUSHING
ESSENTIAL 415V
MCC MAIN SUPPLY

F13-HF-014
RECYCLE CRUSHING
415V MCC
ACTIVE HARMONIC FILTER
30.300A
2000-F13-SCM-EL-10445

t° M 3~ 37kW
F13-FN-402
RECLAIM TUNNEL
DUST COLLECTOR
FAN
2000-F13-SCM-EL-10202

RTD M 3~ 280kW
F13-FA-411
COARSE ORE
RECLAIM FEEDER 1
2000-F13-SCM-EL-10203

RTD M 3~ 280kW
F13-FA-412
COARSE ORE
RECLAIM FEEDER 2
2000-F13-SCM-EL-10204

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-014 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10401.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F13-SLD-EL-10401_Rev2

Recycle crushing F13-SW-014 415V MCC single line diagram sheet 1